The Two-Site Inclusion Process: Complete Hahn Spectrum and the Absorbing Face

HCS-C326 theorem-and-evidence package

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Abstract

We diagonalize the convention-locked two-site symmetric inclusion process. The result includes its beta-binomial law, every Hahn mode and finite-sum norm, the full semigroup kernel, sharp L^2 decay, and the singular zero-parameter absorbing limit. Exact rational grids audit, but do not prove, the theorem.

1 Frozen generator and reversible law

Fix $N \in \mathbb{Z}_{\geq 0}$ and $\alpha > 0$. State $x \in S_N = \{0, \dots, N\}$ is the first-site occupancy; the second is $N - x$. Our unscaled continuous-time generator is

$$(Lf)(x) = b_x[f(x+1) - f(x)] + d_x[f(x-1) - f(x)], \quad b_x = (N-x)(\alpha + x), \quad d_x = x(\alpha + N - x), \tag{1}$$

with unavailable endpoint terms zero. Put

$$\pi(x) = \frac{N!(\alpha)_x(\alpha)_{N-x}}{x!(N-x)!(2\alpha)_N}. \tag{2}$$

Theorem 1. *For $N \geq 1$, (2) is the unique reversible law. Define*

$$H_j(x) = {}_3F_2\left({-j, j+2\alpha-1, -x \atop \alpha, -N}; 1\right), \quad h_j = \sum_{x=0}^{N} \pi(x) H_j(x)^2, \quad 0 \leq j \leq N.$$

Then $h_j > 0$, the H_j are a complete orthogonal basis, and

$$LH_j = -\lambda_j H_j, \qquad \lambda_j = j(j - 1 + 2\alpha).$$

The complete transition kernel is

$$p_t(x, y) = \pi(y) \sum_{j=0}^{N} e^{-\lambda_j t} \frac{H_j(x) H_j(y)}{h_j}. \tag{3}$$

For centered f, $\|e^{tL} f\|_{2,\pi} \leq e^{-2\alpha t} \|f\|_{2,\pi}$; the exponent is sharp. At $\alpha = 0$, the endpoints absorb, absorption is almost sure, and $\mathbb{P}_x(X_\infty = N) = x/N$ for $N \geq 1$. Moreover $\{c\delta_0 + (1-c)\delta_N : 0 \leq c \leq 1\}$ is the complete stationary family, and $\pi_{\alpha,N} \Rightarrow (\delta_0 + \delta_N)/2$ as $\alpha \downarrow 0$. For $N = 0$ the process and law are the singleton.

The adjacent identity $\pi(x) b_x = \pi(x+1) d_{x+1}$ follows by cancelling Pochhammer factors; Chu–Vandermonde gives the denominator in (2). For $N \geq 1$ all interior edge rates are positive, proving irreducibility, uniqueness, and self-adjointness in $L^2(\pi)$.

The evidence grid uses $\alpha \in \{1/2, 1, 3/2, 2\}$ and $0 \leq N \leq 8$: 36 parameter rows, 180 states, and 180 full Hahn vectors. All entries are exact rationals independently reconstructed; finite evidence is not a proof.

Revision certificate: frozen inclusion chain and reversible law

This original round fixes the generator and clock, derives beta-binomial detailed balance, states the complete theorem, and records the exact grid.

References

[1] C. Giardina, F. Redig, and K. Vafayi, Correlation inequalities for interacting particle systems with duality, *J. Stat. Phys.* 141 (2010); DOI: 10.1007/s10955-010-0055-0; arXiv:0906.4664.

[2] NIST Digital Library of Mathematical Functions, Sections 18.19, 18.20.5, and 18.22(ii), Hahn definitions, representation, and difference equations.